Building AI at the Edge



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.NET MAUI Day Cologne
23 October 2026

Welcome to the edge

Where the app meets a person

1 Building with AI

2 Building AI into apps

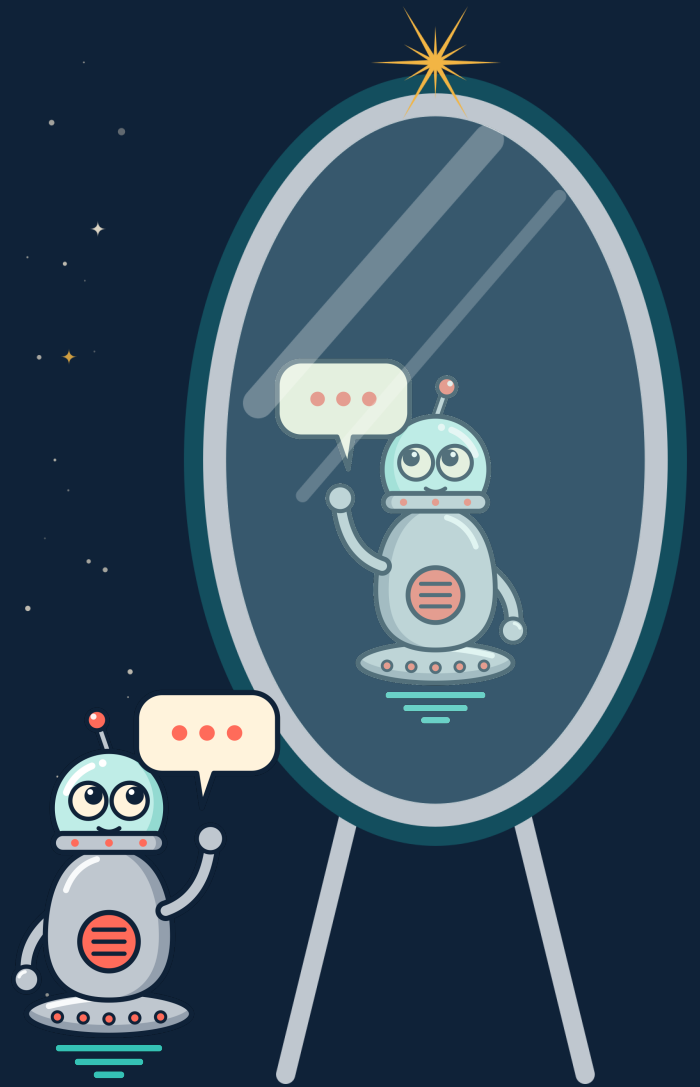


Longer loops need more trust



Garbage in, garbage out

`AGENTS.md`	**Skills**
`copilot-instructions.md`	**MCP servers**
`CLAUDE.md`	**Tools**
Your prompt	**Your existing code**



When it's dumb, check the mirror.

Or it's doing something for reasons you don't understand yet.

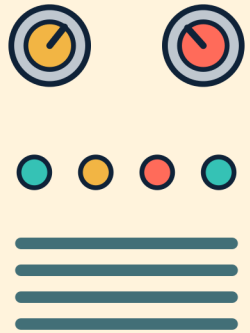
A mirror moment

It looked like a dumb model. **It was our contradiction.**

Sentence Studio's coach, Sam. Techorama rehearsal, October 21

```
> What's due for review?
  Sam: I wasn't able to action that.          3 of 3

Schema:    a window is "for example 7, 14, or 30"
Validator: only 7, 14, or 30. Otherwise refuse.
```



What changed the result

Reasoning effort low: 7 of 7 answered

Stricter wording: 5 of 5 declined

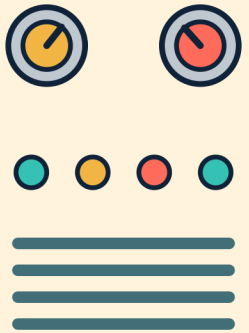


Set the agent up

The `maui` CLI: one front door for you and your agent

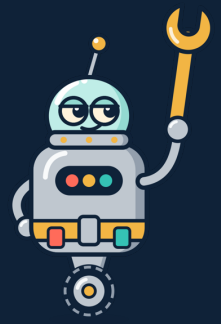
Experimental | dotnet/maui-labs

```
$ maui doctor                # is this machine ready?
$ maui device list --json    # where can I deploy?
$ maui devflow init          # add DevFlow skills to the repo
$ maui devflow mcp           # serve the tools to your agent
$ maui devflow ui tree       # see the running app
```

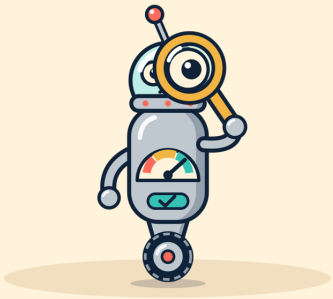


Every command speaks agent

`--json` `--dry-run` `--ci`



Give the agent eyes and hands

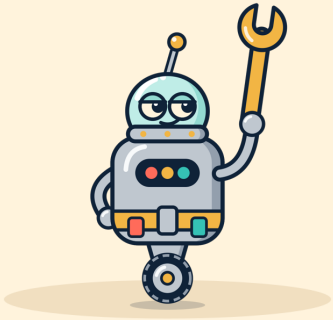


DevFlow

Inside your MAUI app

Tree, tap, type, screenshots, logs, network, Blazor WebView. As MCP tools.

Experimental | maui-labs



Ailoha

Inside any app

The same idea for MAUI, native iOS and Android, Flutter, React Native, WinUI.

Experimental | Redth

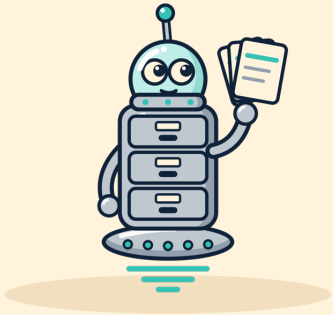


Mobile Canvas

The device you both watch

Live simulators in Copilot and VS Code. Watch the agent's cursor, or take over.

Community | Redth



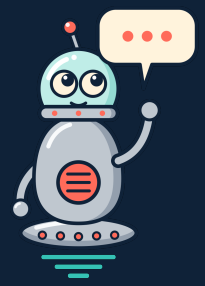
Sherpa

Your environment, sorted

Desktop app for SDKs, emulators, simulators, and signing. Copilot suggests setup fixes.

Community | Redth

Blocked? Fix the tool.

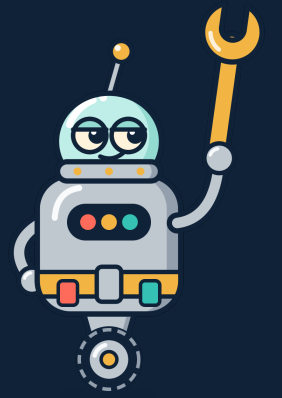


Agent

I can't tap the button on this popup, so I can't dismiss it.

Me

Yes, you can.
And if you really can't, `fix maui devflow`.



It's open source. Ask your assistant to fix it.

A few minutes of tokens now saves hours later.

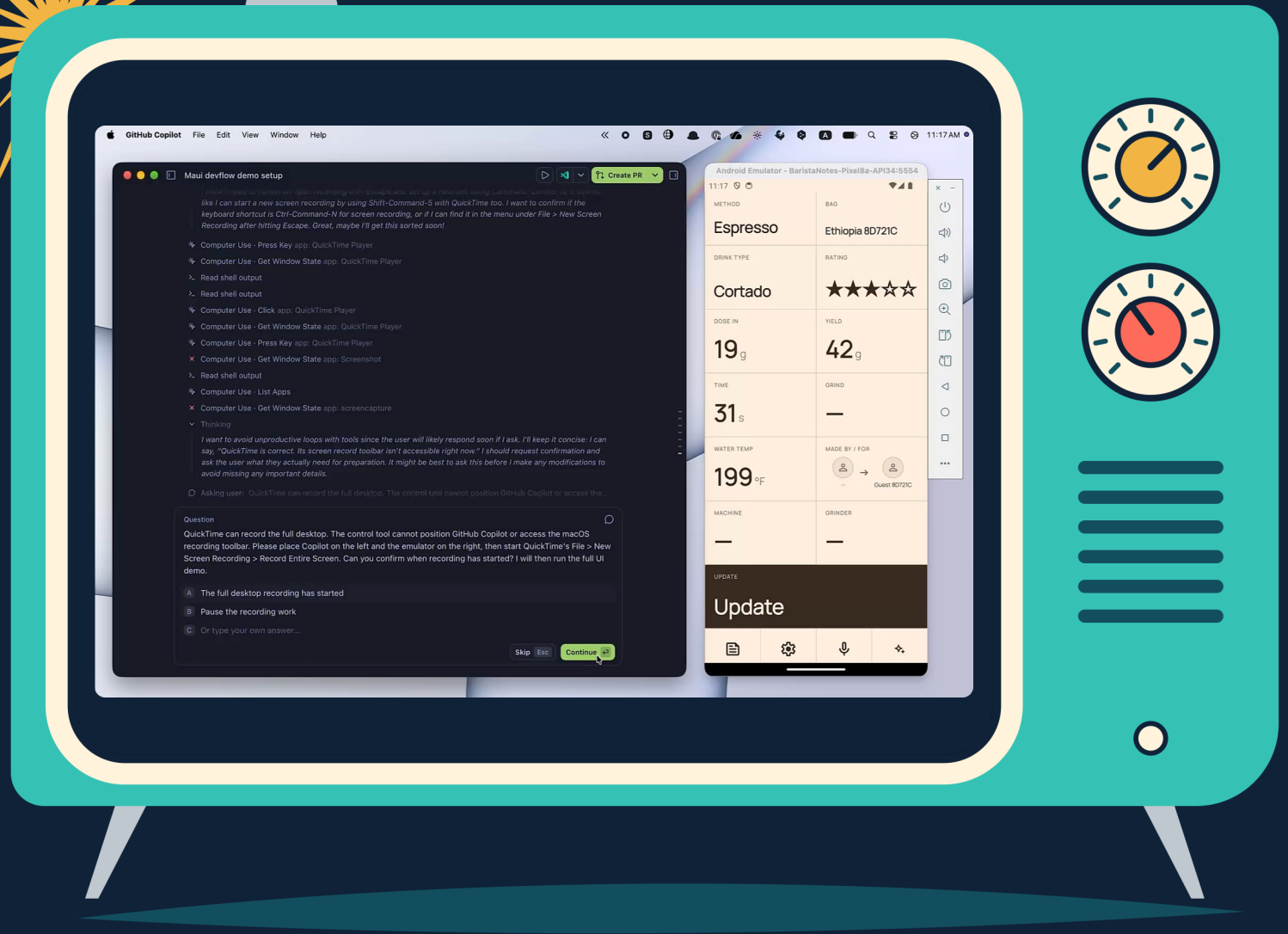
My fixes in dotnet/maui-labs

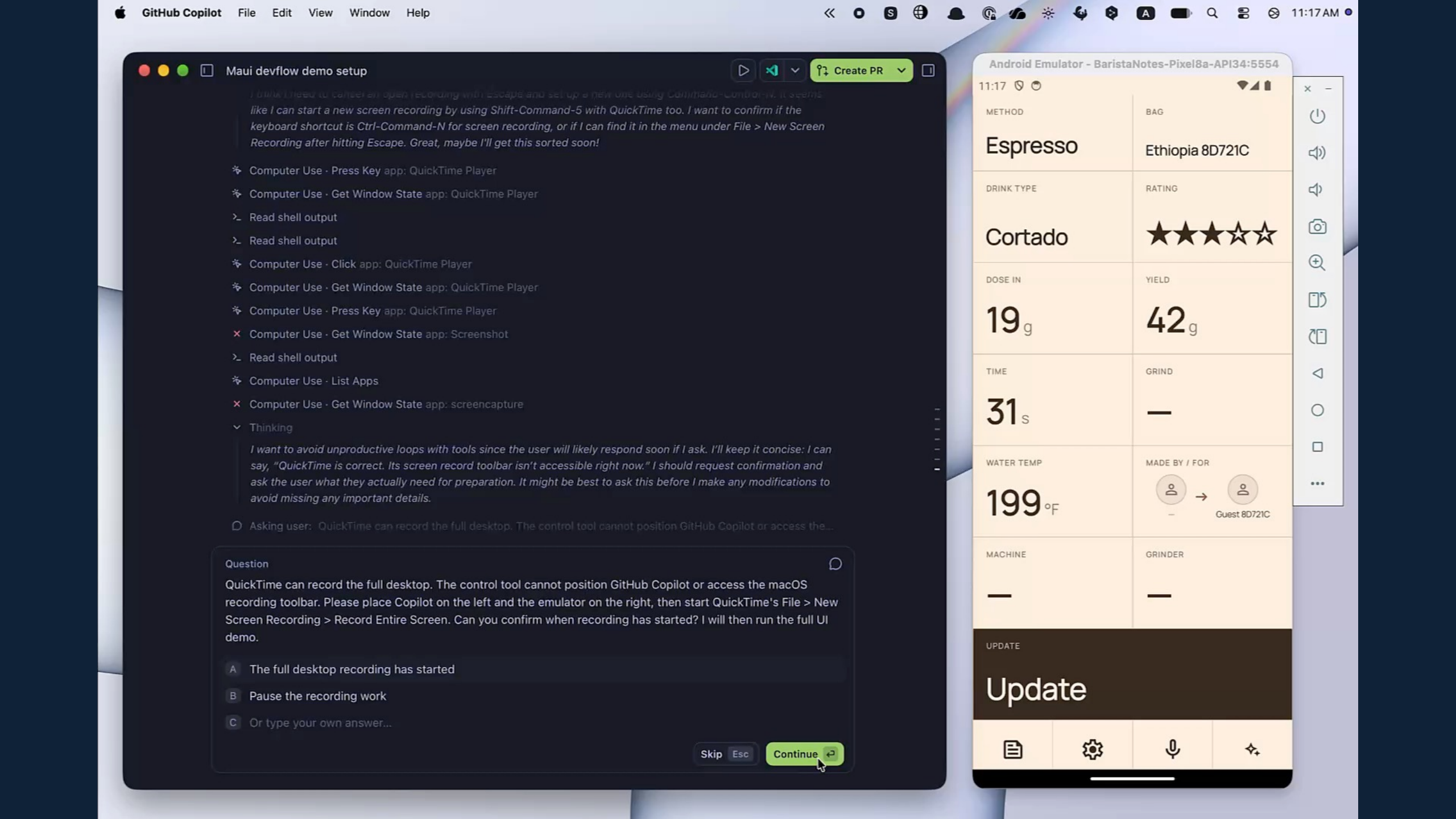
#727 popups **#648 Android taps** **#527 Native AOT** **#65 broker startup**

Demo 1

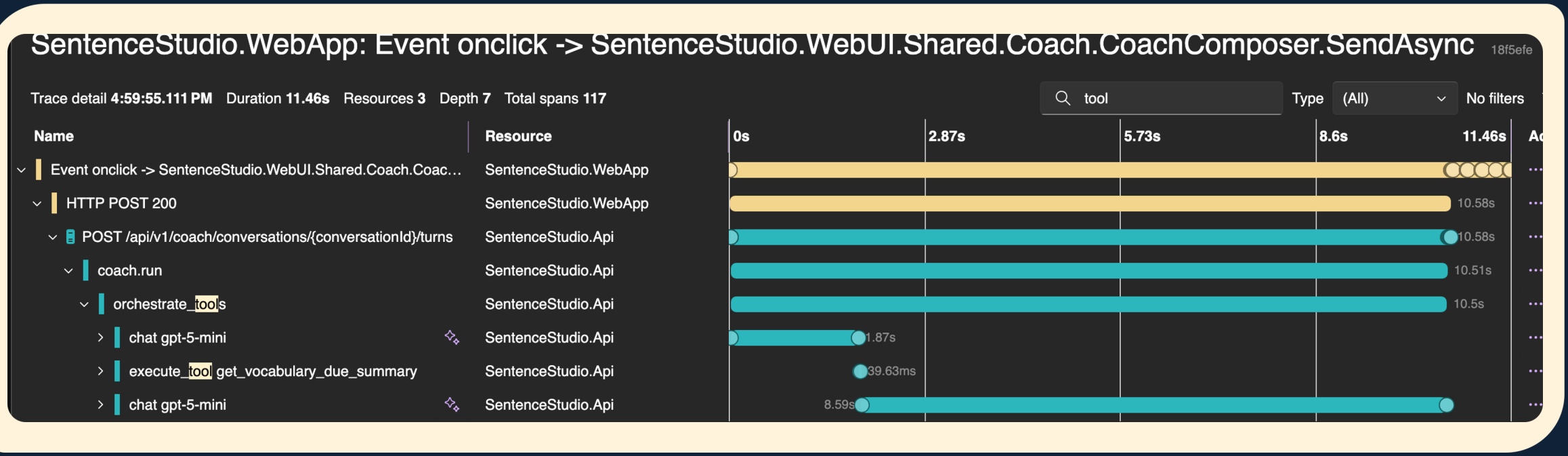
Watch it work

Copilot drives Barista Notes with maui devflow




GitHub Copilot File Edit View Window Help
11:17 AM
Maui devflow demo setup
Create PR
like I can start a new screen recording by using Shift-Command-5 with QuickTime too. I want to confirm if the keyboard shortcut is Ctrl-Command-N for screen recording, or if I can find it in the menu under File > New Screen Recording after hitting Escape. Great, maybe I'll get this sorted soon!
Computer Use · Press Key app: QuickTime Player
Computer Use · Get Window State app: QuickTime Player
Read shell output
Read shell output
Computer Use · Click app: QuickTime Player
Computer Use · Get Window State app: QuickTime Player
Computer Use · Press Key app: QuickTime Player
Computer Use · Get Window State app: Screenshot
Read shell output
Computer Use · List Apps
Computer Use · Get Window State app: screencapture
Thinking
I want to avoid unproductive loops with tools since the user will likely respond soon if I ask. I'll keep it concise: I can say, "QuickTime is correct. Its screen record toolbar isn't accessible right now." I should request confirmation and ask the user what they actually need for preparation. It might be best to ask this before I make any modifications to avoid missing any important details.
Asking user: QuickTime can record the full desktop. The control tool cannot position GitHub Copilot or access the...
Question
QuickTime can record the full desktop. The control tool cannot position GitHub Copilot or access the macOS recording toolbar. Please place Copilot on the left and the emulator on the right, then start QuickTime's File > New Screen Recording > Record Entire Screen. Can you confirm when recording has started? I will then run the full UI demo.
A The full desktop recording has started
B Pause the recording work
C Or type your own answer...
Skip Esc
Continue
Android Emulator - BaristaNotes-Pixel8a-API34:5554
11:17
METHOD
Espresso
BAG
Ethiopia 8D721C
DRINK TYPE
Cortado
RATING
DOSE IN
19 g
YIELD
42 g
TIME
31 s
GRIND
WATER TEMP
199 °F
MADE BY / FOR
Guest 8D721C
MACHINE
GRINDER
UPDATE
Update

Observe the AI inside your app



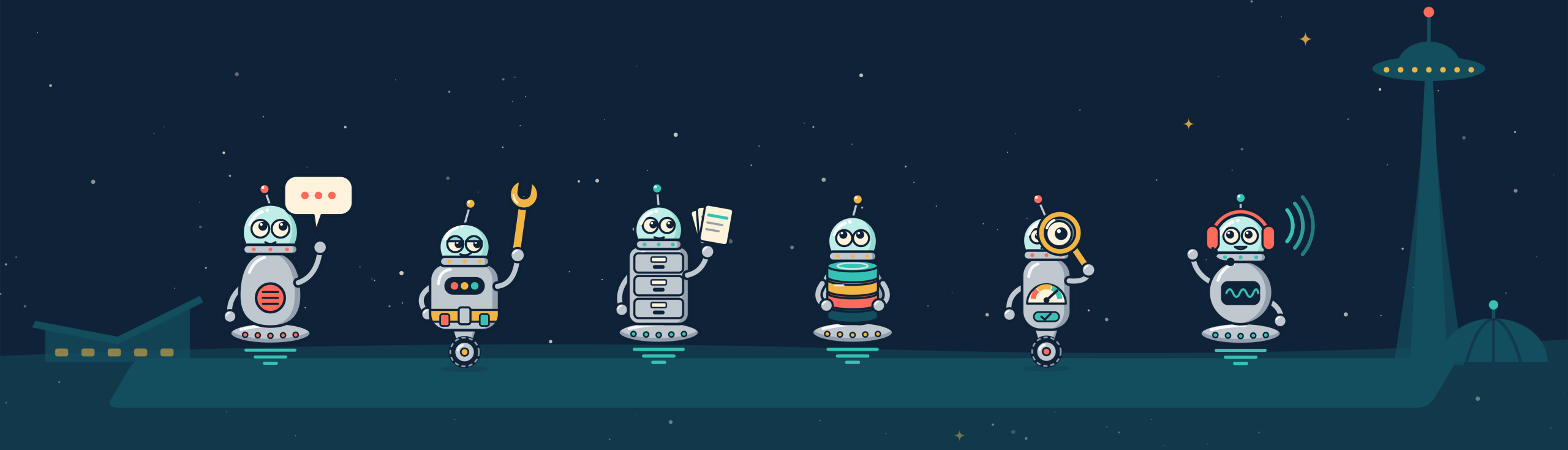
Every model and tool call

Exactly what the model saw

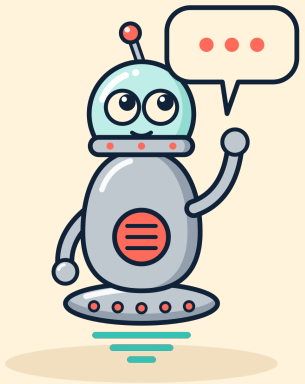
Prompt capture: development only

Sentence Studio: one turn with Sam, traced in the Aspire dashboard

What AI can do for your users

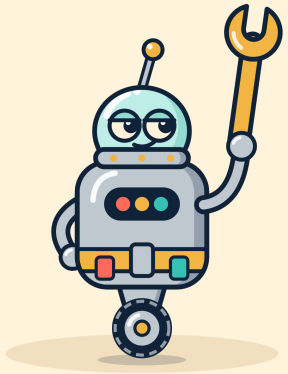


What you told us you want AI to do



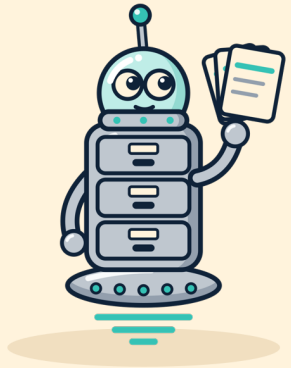
Talk to the app

Voice and chat that run real tasks



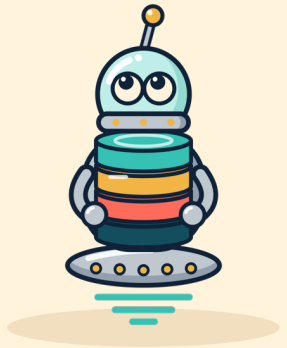
Automate manual work

Forms, orders, data entry



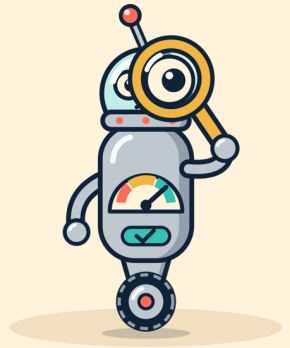
Answer from app knowledge

Docs and systems, then a person



Turn data into decisions

Summaries, alerts, reports



Read documents and images

Invoices, parts, inspections



Personalize the experience

Learning, coaching, stories

October MAUI survey | 194 written answers | themes, not a ranking

At the edge, the model can be here



On the device

Offline | Private | Fast | No per-call cost

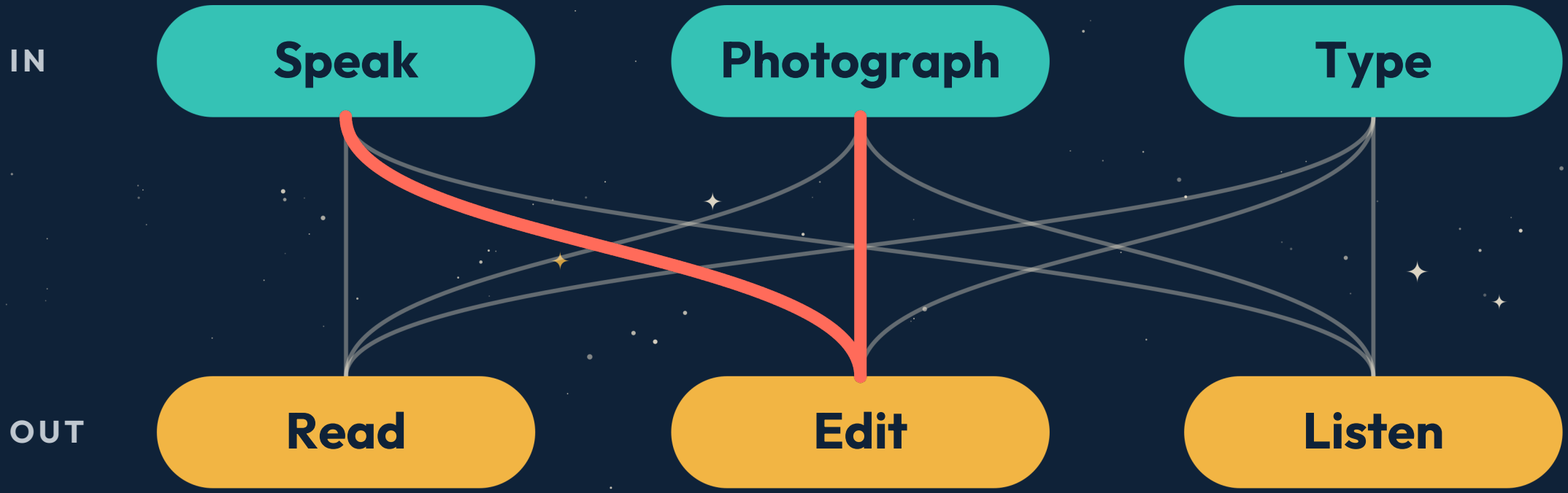
Foundation Models | Windows AI | AICore | ONNX Runtime

Check availability.
Keep a fallback.

In the cloud

Bigger models | Longer reasoning

Choose input and output separately



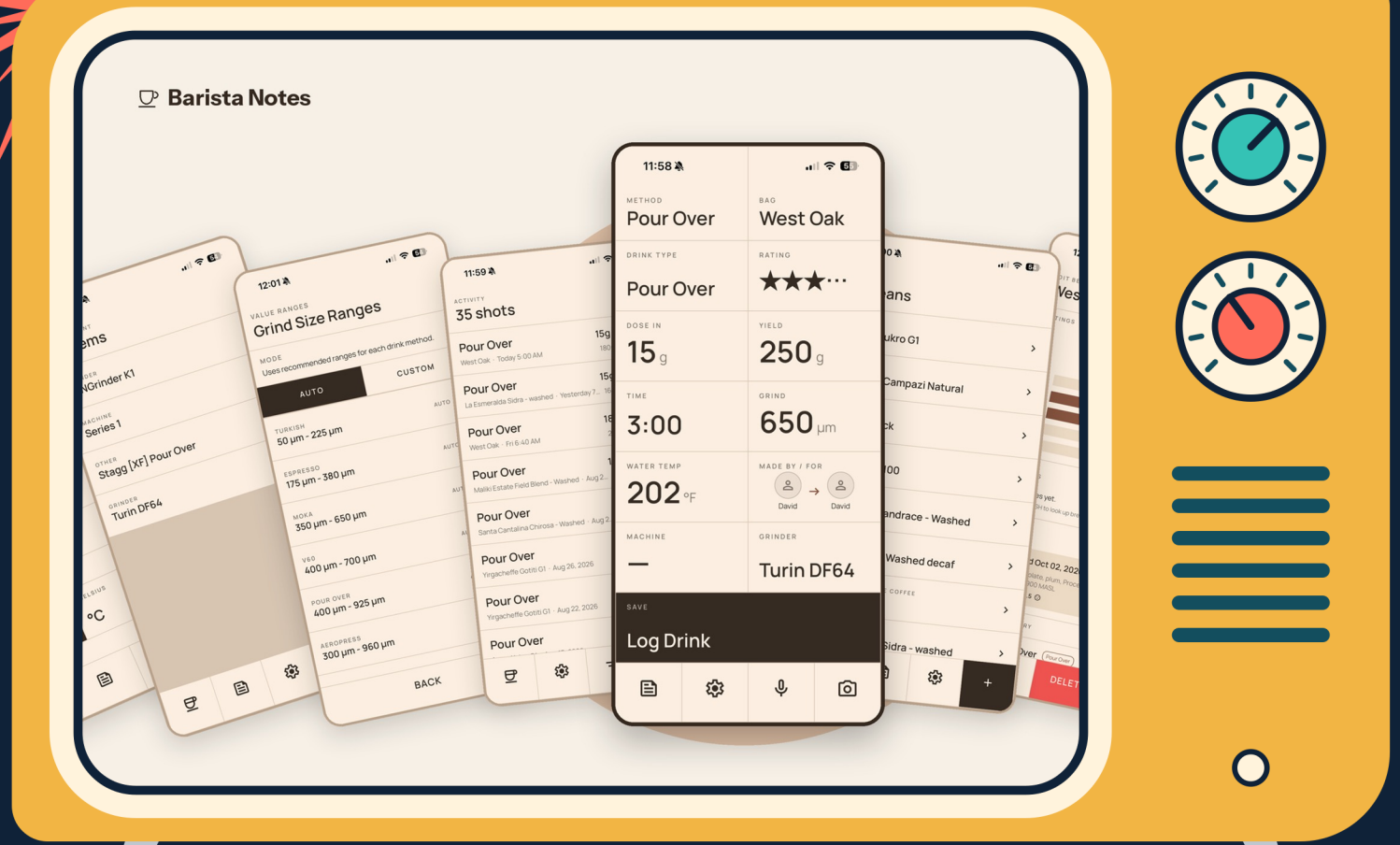
A voice request doesn't need a spoken answer.

A photo needs a form you can check, not a paragraph.

Demo 2

Talk to it. Show it.

Barista Notes: voice commands and photos



Demo 3

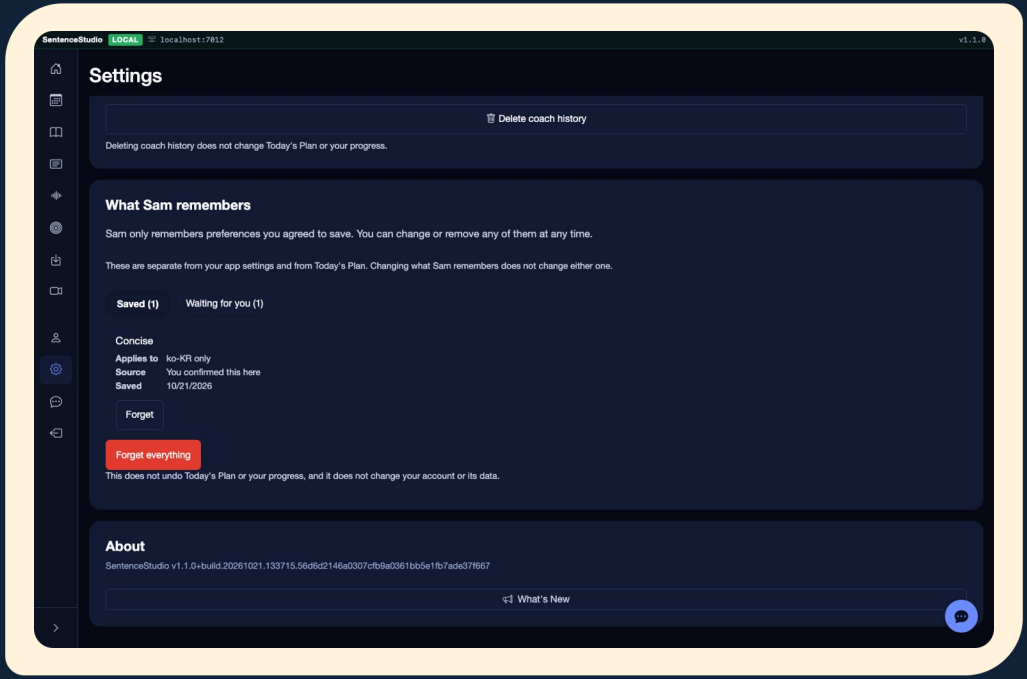
See it.
Say it.

Sentence Studio: scenes, voice, import

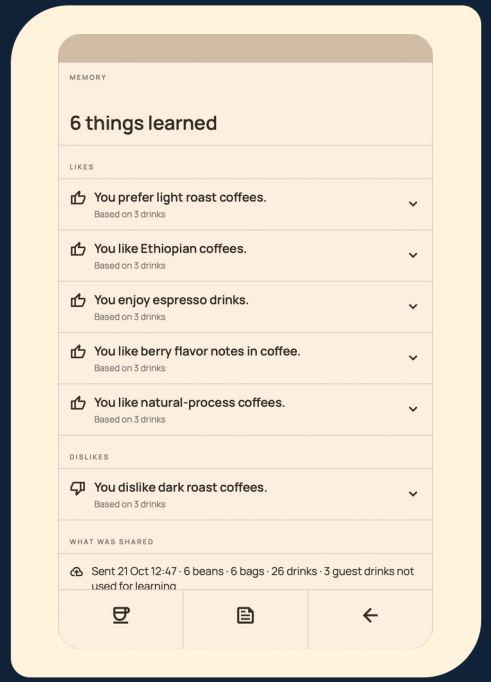


Apps already remember the person

Every preference cites its evidence



Sentence Studio: Sam



Barista Notes



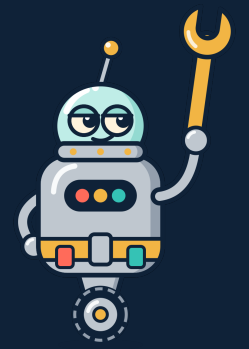
candidate → approved → edited or forgotten



Learn the routine, not just the person

An idea we're exploring, not a product

1. Do the task with the app's tools
2. Watch corrections and check the outcome
3. Keep the steps that repeat
4. Propose a saved routine
5. Reuse with fresh inputs and current permissions



Less repeated discovery.

Take this home

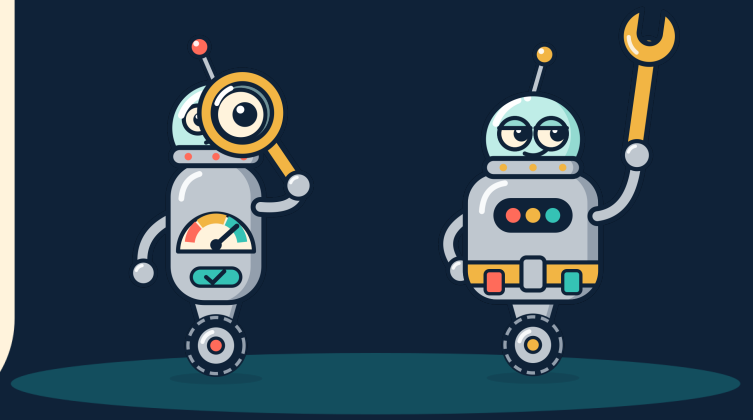
1. **Set your agent up**
   Garbage in, garbage out

2. **Give it eyes and hands**
   And when it can't, fix the tool

3. **Start with work your users repeat**
   Then let the app learn the routine

4. **Choose input and output separately**
   Talk, show, type. Read, edit, listen.





Questions?

What do your users do twice?



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